

Akanksha Dadhich

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EDUCATION

Indian Institute of Technology Bombay: - MTech(CSE)

Year of passing- 2026. CGPA- 9.39

Birla Institute of Technology, Mesra — BE(CSE)

Year of passing- 2021. CGPA- 8.24

Company Experience – 2 years

Programming language: *Java*

Summary : Software Engineer with strong foundations in Java, data structures, and systems programming, currently pursuing an M.Tech in Computer Science at IIT Bombay, previously worked in Amazon as Backend Engineer. Experienced in building backend services, concurrent systems, and Kubernetes-based platforms.

Experience

Backend Engineer/Administrator - Dept. of CSE, IIT Bombay

July 2023 – Present

- Build a **production-grade course enrollment platform** handling 50K+ requests and 1K+ concurrent users during peak admission windows.
- Developed **RESTful microservices using Spring Boot** for enrollment, processing, and status management.
- Designed a **Kafka-based asynchronous enrollment pipeline** with course-id-keyed partitions to guarantee per-course FIFO ordering while enabling parallel processing.
- Used **PostgreSQL as the system of record** with ACID transactions, row-level locking, and unique constraints to prevent race conditions and duplicate enrollments.
- Built a **MongoDB-backed read model** for fast status lookups using event-driven updates after PostgreSQL commits.
- Implemented **JWT authentication**, and enforced **idempotent enrollment workflows** to ensure consistency during retries
- Containerized services with **Docker** and used **Ansible in lab environments** for deploying and managing multiple services.

Amazon

Cloud Backend Engineer

April 2022 – June 2023

- Implemented **API rate limiting using Spring Boot** for **Command Center**, an internal support tool, protecting downstream services from **traffic spikes of 5–10× during incident scenarios**.
- Designed **scalable backend services** enforcing **per-user limits (~100–300 requests/min)**, ensuring fair usage across **50+ internal support engineers**.
- Improved **system reliability** by reducing **request overload incidents** by ~30–40% during peak support activity
- Added **monitoring and alerts** to track **throttled requests, and latency**.
- Owned the **Command Center backend**, collaborating with support and engineering teams to tune limits without impacting **critical workflows**

ZS Associates

Technology Solutions Associate

Feb 2021 – Nov 2021

- Created optimized queries to improve performance, data correctness and developed SQL-backed internal dashboards

PROJECTS

Sankshpika: Legal Document Summarizer

Tech used : Python, PyTorch, OpenAI API, TF-IDF, LLM

- Finetuned LegalLED with Indian Dataset while extracting important sentences using the TF-IDF method.
- Combined the summaries with OpenAI overcoming token restrictions achieving a ROUGE score of 0.45.

Concurrent Auto Grading Server

Tech used C, Socket Programming, Multithreading

- Engineered a multi-client autograding server in C using socket programming, supporting Normal, Multithreaded, Thread Pool, and Asynchronous execution modes.
- Evaluated concurrency strategies to ensure correctness and isolation under parallel submissions.

Kubernetes-based Platform for Hosting Isolated

User Services Tech Used: Kubernetes, Docker

- Designed and implemented a Kubernetes-based platform to deploy user-provided static websites as isolated Deployments and Services.
- Exposed services externally using Kubernetes NodePort, enabling independent access and basic load balancing.
- Automated deployment & service configuration using custom scripts, reducing manual setup time.

TECHNICAL SKILLS

- Backend & Systems: Spring Boot, REST APIs, Multithreading, Concurrency, Linux
- Distributed & Cloud: AWS (EC2, S3, MWAA), Kubernetes, Docker
- CS Foundations: Data Structures, Algorithms, Operating Systems, Computer Networks
- Tools: GitHub, Bash
- Programming Languages: Java (Primary), C, Python, SQL

CERTIFICATIONS

AWS certifications: id: 0GJK3WG2YJ441E3P

AWS Certified Solutions Architect – Associate (2022)

Linux RedHat Certificate:

Red Hat Certified System Administrator

(RHCSA)(EX200)(2019)

ACHIEVEMENTS

- Secured 99.39 percentile in GATE CS 2023
- Received Research Assistantship Excellence Award for outstanding work (Spring 2024).